

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
 THIRD YEAR(Computer Science)
 SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours

Batch 2023

Dated : 03-JUN-26

Max Marks : 60

Operating Systems - CT-353

Question 1

[C4 – 12 marks]

- a) The following processes arrive in a system with their arrival times, burst times, and priorities.

Process	Arrival Time (AT)	Burst Time (BT)	Priority
P1	0	7	2
P2	2	4	1
P3	4	1	3
P4	5	4	2
P5	6	6	4

10-2 5-8
 4-8 (4,6,5,2)
 9-2 6-4
 10-2 11-4 (13,4,9)

- Draw four Gantt charts showing the execution of the given processes using FCFS, SJF (non-preemptive), non-preemptive priority scheduling, and Round Robin with a time quantum of 4ms.
- For each algorithm, calculate the average turnaround time and average waiting time across all processes

Question 2

[C4 – 12 marks]

- Explain how a modern computer works by describing the Von Neumann architecture with a diagram, highlighting its main components and working principle. (6)
- Compare the following concepts in detail: (6)
 - Single-threaded process vs Multi-threaded process → 2
 - Symmetric Multiprocessing (SMP) vs Asymmetric Multiprocessing (AMP)
 - Symmetric clustering vs Asymmetric clustering.

Question 3

[C4 – 12 marks]

Analyze the system with 5 processes (P0–P4) and 4 resource types (A,B,C,D) at time T0:

Process	Allocation				Max				Current Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	1	1	0	0	2	1	0	1	5	2	0
P1	1	2	3	1	1	6	5	2				
P2	1	3	6	5	2	3	6	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

1 4 4 1

[3, 17, 16, 12]

- (a) Use the Banker's Algorithm to determine whether the system is in a safe state. If the system is safe, give the safe sequence.

- (b) For each of the following requests, determine whether the request can be granted

- P1 requests (2, 1, 1, 0)
- P1 requests (0, 2, 1, 0)

6

0 3 4 1 2

Question 4

[C2 – 12 marks]

- a) Define the following terms:
i. OpenMP thread, ii. task_struct, iii. Priority Inversion, iv. Lightweight Process ² (4)
- b) Explain the working of the Readers-Writers Problem in operating systems. Illustrate your answer using a suitable execution sequence (traversal) showing the interaction between a reader and a writer, where a writer must wait while a reader is accessing the shared resource. (4)
- c) Explain how an operating system kernel handles process execution in a critical system. Also discuss the solutions to the Critical-Section Problem. (4)

Question 5

[C2 – 12 marks]

- a) Compare thread models in Windows and Linux operating systems. Explain the Windows one-to-one thread model and thread context, including the roles of ETHREAD, KTHREAD, and TEB (use a diagram). Also state key differences between Windows and Linux thread implementations. (6)
- b) Explain POSIX threads (Pthreads) in C programming, including the key thread functions used for thread creation and management. Also differentiate between implicit and explicit threading, and state which approach is more suitable for programmers with proper justification. (6)